DEEP LEARNING FOR INSTANCE SEGMENTATION AND QUANTIFICATION OF BEAD DEFECTS IN SEM IMAGES OF ELECTROSPUN NANOFIBRE MATS

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ERLIND SKURA

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Approval sheet of the Thesis

This is to certify that we have read this thesis entitled "**DEEP LEARNING FOR INSTANCE SEGMENTATION AND QUANTIFICATION OF BEAD DEFECTS IN SEM IMAGES OF ELECTROSPUN NANOFIBRE MATS**" and that in our opinion it is fully adequate, in scope and quality, as a thesis for the degree of Master of Science.

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Dr. Florenc Skuka
Head of Department
Date: September 2026

Examining Committee Members:

Assoc. Prof. Dr. Arban Uka	(Computer Engineering)	__________
	(Computer Engineering)	__________
	(Computer Engineering)	__________

I hereby declare that all information in this document has been obtained and presented in accordance with academic rules and ethical conduct. I also declare that, as required by these rules and conduct, I have fully cited and referenced all material and results that are not original to this work.

Name Surname: Erlind Skura

Signature: ____________________

ABSTRACT

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Skura, Erlind

M.Sc., Department of Computer Engineering

Supervisor: Assoc. Prof. Dr. Arban Uka

Electrospinning produces nanofibre mats for filtration, wound dressing and tissue scaffolding, but the process is prone to a characteristic defect in which polymer solution collects into droplets along the fibres. These bead defects degrade porosity and mechanical strength, so quantifying them is a routine part of process optimisation. In practice the quantification is done by hand: an operator opens each scanning electron microscopy (SEM) micrograph in ImageJ and outlines beads individually, which is slow and difficult to reproduce.

This thesis develops an automated instance segmentation pipeline for bead defects in SEM micrographs of electrospun nanofibre mats. The dataset consists of 11 micrographs of four specimens, each imaged at 500×, 1000× and 3000× magnification, containing 606 manually delineated beads. Because the beads span a sixfold range in apparent size and the four specimens are the only independent units in the data, the evaluation uses leave-one-specimen-out cross-validation, which prevents the same specimen from appearing in both the training and the test partition. Mask R-CNN is compared against a U-Net semantic baseline and against a classical Otsu-plus-watershed pipeline representing current manual practice. Beyond detection quality, the pipeline reports the quantities the laboratory actually uses: bead count, bead areal density, and the bead size distribution in physical units, obtained by calibrating each micrograph against its burned-in scale bar.

[Results sentence to be written after Chapter 5: report detection AP, counting error against the manual ground truth, and agreement between the predicted

and manual size distributions.]

iv

Keywords: *Instance Segmentation, Electrospun Nanofibres, Bead Defects, Scanning Electron Microscopy, Mask R-CNN*

ABSTRAKT

MËSIMI I THELLË PËR SEGMENTIMIN E INSTANCAVE DHE KUANTIFIKIMIN E DEFEKTEVE TË TIPIT RRUAZË NË IMAZHE SEM TË MEMBRANAVE NANOFIBROZE TË ELEKTROTJERRURA

Skura, Erlind
Master Shkencor, Departamenti i Inxhinierisë Kompjuterike
Udhëheqësi: Assoc. Prof. Dr. Arban Uka

Elektrotjerrja prodhon membrana nanofibroze që përdoren në filtrim, në fasho plagësh dhe si skela për inxhinierinë e indeve, por procesi është i prirur ndaj një defekti karakteristik, ku tretësira polimere mblidhet në pika të vogla përgjatë fibrave. Këto defekte të tipit rruazë ulin poroziteten dhe qëndrueshmërinë mekanike, prandaj matja e tyre është pjesë e zakonshme e optimizimit të procesit. Në praktikë kjo matje kryhet manualisht: operatori hap secilin imazh të mikroskopisë elektronike me skanim (SEM) në ImageJ dhe vizaton konturet e rruazave një nga një, gjë që është e ngadaltë dhe e vështirë për t'u riprodhuar.

Kjo tezë ndërton një sistem të automatizuar për segmentimin e instancave të defekteve të tipit rruazë në imazhe SEM të membranave nanofibroze. Grupi i të dhënave përbëhet nga 11 imazhe të katër mostrave, secila e fotografuar me zmadhim 500×, 1000× dhe 3000×, me 606 rruaza të anotuara manualisht. Duke qenë se rruazat shfaqen me përmasa që ndryshojnë deri në gjashtë herë dhe se katër mostrat janë njësitë e vetme të pavarura në të dhëna, vlerësimi kryhet me validim të kryqëzuar duke lënë jashtë një mostër të plotë (*leave-one-specimen-out*), çka pengon shfaqjen e së njëjtës mostër njëkohësisht në trajnim dhe në testim. Mask R-CNN krahasohet me një model bazë U-Net dhe me një metodë klasike Otsu me *watershed*, e cila përfaqëson praktikën aktuale manuale. Përveç cilësisë së detektimit, sistemi raporton madhësitë që laboratori i përdor vërtet: numrin e rruazave, dendësinë e tyre për njësi sipërfaqeje dhe shpërndarjen e madhësisë në njësi fizike, e llogaritur duke kalibruar secilin imazh me shiritin e shkallës të printuar brenda tij.

[Fjalia e rezultateve shkruhet pas Kapitullit 5.]

Fjalë kyçe: *Segmentim Instancash, Nanofibra të Elektrotjerrura, Defekte Rruazë, Mikroskopi Elektronike me Skanim, Mask R-CNN*

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LIST OF ABBREVIATIONS

AJI	Aggregated Jaccard Index
AP	Average Precision
CNN	Convolutional Neural Network
EHT	Extra High Tension (accelerating voltage)
FPN	Feature Pyramid Network
GPU	Graphics Processing Unit
IoU	Intersection over Union
LOSO	Leave-One-Specimen-Out
mAP	Mean Average Precision
NMS	Non-Maximum Suppression
PQ	Panoptic Quality
R-CNN	Region-based Convolutional Neural Network
RoI	Region of Interest
RPN	Region Proposal Network
SAM	Segment Anything Model
SEM	Scanning Electron Microscopy
WD	Working Distance

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CHAPTER 1

INTRODUCTION

1.1 Introduction

Electrospinning draws a polymer solution into fibres with diameters between tens of nanometres and a few micrometres by applying a high electric field between a spinneret and a grounded collector. The resulting non-woven mats combine a very high surface-to-volume ratio with an interconnected pore structure, which is why they are used in air and water filtration, in wound dressings, in drug delivery and as scaffolds for tissue engineering.

The process is not, however, uniformly well behaved. When the electrostatic stretching force is insufficient to overcome the surface tension of the jet, the solution collapses locally into droplets that solidify as ellipsoidal swellings along the fibre. These *bead defects*, first characterised systematically by Fong, Chun and Reneker [5], are the standard indicator that the spinning parameters are outside their usable window. Beads reduce the effective surface area of the mat, disturb its pore-size distribution and act as stress concentrators that weaken it mechanically. Counting and measuring them is therefore a routine step whenever an electrospinning recipe is being tuned.

That step is currently manual. An operator loads each scanning electron microscopy (SEM) micrograph into ImageJ [19], identifies the beads by eye and outlines them one at a time. A single micrograph in the dataset used here contains up to 152 beads, and each of them has to be traced individually. The procedure is slow, it does not scale to the number of micrographs that a parameter study produces, and because the boundary between a large bead and a locally thick fibre segment is a matter of judgement, it is difficult to reproduce between operators or even between sessions.

1.2 Background and Motivation

Automating this measurement is an image segmentation problem, but not the easiest kind. Three properties of SEM micrographs of nanofibre mats make it harder than the generic case.

First, the beads sit on top of a dense, high-contrast background of overlapping fibres. The fibres themselves are bright, elongated and cross each other at every angle, so simple intensity thresholding cannot separate beads from background: both occupy similar grey levels, and the texture energy of the fibre mesh is comparable to that of a bead surface.

Second, the objects of interest touch and overlap. Beads frequently occur in clusters along the same fibre, and a method that produces only a foreground mask will merge such a cluster into a single connected component. Since the quantity the laboratory needs is a *count* and a *size distribution*, merging two beads into one is not a small error: it changes the reported count and it invents a non-existent large particle in the size histogram. This is what makes the problem an instance segmentation problem rather than a semantic one.

Third, the data are scarce and are acquired at several magnifications. The dataset available for this thesis contains 11 annotated micrographs drawn from four specimens, each imaged at 500×, 1000× and 3000×. A bead that is about 14 pixels across at 500× is about 76 pixels across at 3000× – the same physical object with a sixfold difference in apparent size. A model trained on this data has to be scale-robust, and the evaluation protocol has to account for the fact that the four specimens, not the eleven images, are the independent units of the experiment.

Modern instance segmentation architectures address the first two difficulties directly. Mask R-CNN [6] detects each object separately and predicts a mask within its own region proposal, so touching beads remain separate instances by construction. The third difficulty – little data, evaluated honestly – is what determines whether the reported numbers mean anything, and it receives explicit attention in this thesis.

1.3 Problem Statement

This thesis addresses the automated detection, delineation and quantification of bead defects in SEM micrographs of electrospun nanofibre mats. Four questions are posed:

1. Can a deep instance segmentation model delineate individual bead defects in SEM micrographs of nanofibre mats accurately enough to replace manual anno-

tation, when only 11 annotated micrographs from four specimens are available?

2. Does instance segmentation offer a measurable advantage over semantic segmentation for this task, specifically in separating touching beads and therefore in the accuracy of the reported bead count?

3. How does a deep model compare against the classical image-processing pipeline (thresholding followed by watershed splitting) that represents current laboratory practice?

4. Do the derived physical quantities – bead count, areal density and size distribution in micrometres – agree with those obtained from the manual annotation closely enough to be used in process optimisation?

The scope is restricted to bead defects in micrographs of a single material system, imaged on one instrument. Fibre diameter measurement, three-dimensional structure and the relationship between spinning parameters and bead formation lie outside the scope, although Section 6.4 returns to them.

1.4 Research Objectives

The objectives are stated below with the criteria that will be used to judge whether each has been met.

- **Dataset characterisation and calibration.** Establish what the available data actually contain and place every measurement on a physical scale. This includes recovering the pixel size of each micrograph from its burned-in scale bar and verifying the calibration by checking that the bead size distribution is consistent across the three magnifications. Success criterion: median bead diameter agreeing to within 20% across 500×, 1000× and 3000×.

- **A leakage-free evaluation protocol.** Define a partitioning scheme appropriate to a dataset whose 11 images derive from only 4 specimens, and demonstrate quantitatively why the naive alternative – a random split over images, or over the augmented copies – overstates performance. Success criterion: a reported gap between the naive and the specimen-wise protocol on the same model.

- **Instance segmentation of beads.** Train Mask R-CNN to delineate individual beads under leave-one-specimen-out cross-validation. Success criterion: to be fixed with the supervisor, provisionally $AP_{50} \geq 0.70$ averaged over the four folds.

- **Comparison against semantic and classical baselines.** Quantify what instance-level modelling buys, by comparing against a U-Net [18] semantic baseline with connected-component labelling and against an Otsu [16] plus watershed [23] pipeline. Success criterion: a clear ranking on counting error, with the merge behaviour of each method reported explicitly.

- **Physical quantification.** Report bead count, areal density (beads mm^{-2}) and the size distribution in micrometres, and compare each against the manual ground truth. Success criterion: counting error and a distributional comparison of predicted versus manual bead diameters.

1.5 Contributions

1. A characterised and physically calibrated dataset of 606 manually delineated bead defects across 11 SEM micrographs of four electrospun specimens, with per-magnification pixel sizes recovered from the instrument's own scale bars and validated for internal consistency.

2. An evaluation protocol for this class of small, hierarchically structured microscopy datasets, in which the specimen rather than the image is the unit of partitioning, together with a measurement of how much a naive split inflates the apparent performance.

3. A controlled comparison of instance segmentation, semantic segmentation and classical thresholding on the same micrographs under the same protocol, evaluated both by segmentation metrics and by the counting and sizing errors that determine whether the method is usable in the laboratory.

4. An open implementation covering annotation conversion, scale calibration, training, evaluation and the reporting of physical quantities.

1.6 Thesis Organization

Chapter 2 reviews electrospinning and bead formation, classical image analysis for fibrous microscopy, and deep segmentation architectures, and identifies the gap this thesis addresses.

Chapter 3 presents the methodology: the dataset and its structure, scale calibration, the specimen-wise evaluation protocol, the three methods under comparison, and the metrics.

Chapter 4 describes the implementation, including annotation conversion, the handling of the burned-in information banner, the augmentation strategy and the training setup.

Chapter 5 reports the experimental results and discusses them.

Chapter 6 concludes and sets out directions for future work.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction

This chapter reviews the three bodies of work that the thesis draws on: the materials-science literature on electrospinning and bead formation, which defines what is being measured and why; the image-analysis literature on fibrous microscopy, which describes how the measurement is currently made; and the computer-vision literature on segmentation, which supplies the methods used here. The chapter closes by stating the gap between them.

2.2 Theoretical Background

2.2.1 Electrospinning and the Formation of Bead Defects

In electrospinning, a polymer solution is fed through a spinneret held at a high potential relative to a collector. Above a threshold voltage the electrostatic repulsion of surface charge overcomes the surface tension of the pendant droplet, which deforms into a Taylor cone and emits a charged jet. The jet undergoes bending instability, stretches by orders of magnitude, and the solvent evaporates, leaving a solid fibre deposited as a randomly oriented non-woven mat [3, 9].

The morphology of the deposit depends on a balance of forces. Surface tension acts to minimise the surface area of the jet and therefore favours the break-up of a cylindrical thread into droplets; viscoelastic stress and charge repulsion oppose this. When the solution has low viscosity, low charge density or high surface tension, the instability is not fully suppressed and the jet partially collapses, producing ellipsoidal swellings distributed along the fibres. Fong, Chun and Reneker [5] established the now-standard account of this behaviour, showing that higher solution viscosity and higher net charge density produce fewer and more elongated beads, and that increasing surface tension has the opposite effect. Bhardwaj and Kundu [1] review how the same parameters – concentration, conductivity, applied voltage, flow rate and tip-to-collector distance – interact in practice.

Because bead content responds so directly to these parameters, bead count and bead size are used as the diagnostic output of a parameter sweep: a recipe is refined until the mat is bead-free, or until the bead population is acceptably small. Beads matter physically as well as diagnostically. They reduce the specific surface area that makes nanofibre mats useful for filtration and catalysis, they perturb the pore-size distribution that governs permeability, and they act as stress concentrators that lower tensile strength.

2.2.2 Scanning Electron Microscopy of Nanofibre Mats

SEM is the standard characterisation tool for these materials. It resolves features well below the diffraction limit of light microscopy and produces images whose contrast derives largely from surface topography, which suits the assessment of fibre morphology. Micrographs are typically acquired at several magnifications: a low magnification to sample enough fibres for a representative measurement, and a higher one to resolve individual features. Instrument software burns an information banner into the image, recording the scale bar, accelerating voltage, working distance and magnification. The banner is convenient for a human reader and is the means by which pixel measurements can be converted to physical units, but it is not part of the specimen and must be excluded before any automated analysis.

2.2.3 Semantic and Instance Segmentation

Semantic segmentation assigns a class label to every pixel; instance segmentation additionally separates the pixels belonging to distinct objects of the same class. The distinction is decisive here. A semantic model that labels every bead pixel correctly still yields a single connected component for a cluster of touching beads, so the derived count is too low and the derived size distribution contains a spurious large object. Since the laboratory endpoint is a count and a size distribution rather than an area fraction, instance-level output is required.

2.3 Related Work

2.3.1 Classical Image Analysis for Fibrous Microscopy

The established approach in the materials literature is a sequence of hand-designed operations: contrast normalisation, thresholding, morphological cleaning and a splitting step. Otsu's method [16] chooses a global threshold by maximising between-class variance and remains the default in tools such as Fiji [19]. Touching objects are then separated by the watershed transform [23], usually applied to the distance transform of the binary mask.

For electrospun materials specifically, DiameterJ [8] provides a validated open-source pipeline for fibre diameter and porosity from SEM micrographs, and is widely used. Its scope, however, is the fibre network: it characterises diameters and intersections rather than isolating discrete defects.

These pipelines have real advantages – they are transparent, they require no training data and their failure modes are interpretable – but they depend on the foreground being separable by intensity. In micrographs of a dense fibre mat this assumption does not hold: bright fibres cross the field at every angle and occupy the same grey levels as bead surfaces, so a global threshold cannot isolate the beads. Watershed splitting also has a well-known tendency to over-segment on noisy distance transforms, which is the failure that most directly corrupts a count.

2.3.2 Encoder–Decoder Networks for Semantic Segmentation

U-Net [18] introduced the encoder–decoder architecture with skip connections that has dominated biomedical segmentation since. The contracting path captures context while the expanding path recovers spatial resolution, and the skip connections restore the high-frequency detail lost to pooling. It was designed explicitly for the regime of interest here, having been demonstrated on training sets of a few dozen annotated images with heavy augmentation. Falk et al. [4] later packaged it for cell counting and morphometry, confirming that the architecture transfers to quantification tasks.

U-Net remains a semantic model, and the standard remedy for instance separation – predicting object boundaries as an auxiliary class and using them to split the foreground – is effective but indirect, and its accuracy depends on how reliably thin boundaries can be predicted.

2.3.3 Instance Segmentation

Faster R-CNN [17] established the two-stage detection paradigm: a region proposal network scores candidate boxes, and a second stage classifies and refines them. Mask R-CNN [6] extends it with a parallel branch that predicts a binary mask inside each proposal, and replaces the coarse RoI pooling with RoIAlign to preserve spatial correspondence. Because each object is handled in its own region, touching objects are separated by construction rather than by post-processing – the property that makes it the natural choice for counting tasks. Feature pyramid networks [14] supply the multi-scale representation that lets a single model handle objects of very different apparent size, which is directly relevant to a dataset spanning a sixfold magnification range.

In microscopy, several methods exploit the shape regularity of the targets. StarDist [20] represents each object as a star-convex polygon, which is well matched to roundish, convex objects and degrades gracefully when they touch. Cellpose [22] learns spatial gradient flows towards object centres and follows them to recover instances, and generalises across cell types without retraining. Bead defects are approximately ellipsoidal and convex, so these shape priors are a plausible fit and are noted here as alternatives.

2.3.4 Segmentation from Few Annotated Images

Training on a handful of annotated micrographs is standard in microscopy, and the countermeasures are well established: transfer learning from large natural-image corpora such as ImageNet [2] or COCO [15], and aggressive geometric and photometric augmentation [21]. Microscopy is more forgiving than most domains in this respect, because flips and rotations are genuinely label-preserving: unlike photographs of natural scenes, a micrograph has no canonical orientation.

Augmentation also introduces a hazard that is easy to overlook. If augmented copies are generated before the data are partitioned, transformed versions of the same source image can appear in both the training and the test set. The test set then measures memorisation rather than generalisation. The same concern applies more broadly to any hierarchical dataset in which several images derive from one physical specimen, a point returned to in Chapter 3.

2.3.5 Foundation Models for Segmentation

The Segment Anything Model [12] is trained on a very large mask corpus and produces class-agnostic masks from point or box prompts, with strong zero-shot transfer. For a domain with only 11 annotated images this is attractive, but SAM has no notion of what a bead is: prompted densely on a fibre mat it will segment fibres, inter-fibre voids and beads indiscriminately. Its realistic role here is as an annotation aid or as a mask proposal generator, not as a standalone detector.

2.4 Evaluation of Instance Segmentation

Detection quality is conventionally summarised by average precision computed over intersection-over-union thresholds, following the COCO protocol [15]. For microscopy, metrics that penalise merges and splits explicitly are more informative: the aggregated Jaccard index [13] accumulates intersection and union over matched objects and charges unmatched ones to the denominator, and panoptic quality [11] factors performance into a recognition and a segmentation term. Since the laboratory endpoint is a count and a size distribution, these segmentation metrics are reported here alongside the counting error and a direct comparison of the predicted and manual size distributions.

2.5 Research Gap

Three gaps emerge from the above.

First, the image-analysis tooling available to the electrospinning community is aimed at the fibre network – diameter, orientation, porosity – rather than at discrete defects. DiameterJ and comparable tools do not count beads, and the quantification of bead defects is consequently still performed by hand.

Second, where deep segmentation has been applied to fibrous microscopy it has usually been semantic, reporting pixel-level overlap. For a task whose output is a count, pixel overlap is the wrong figure of merit: it does not distinguish a model that separates two touching beads from one that merges them, even though only the former produces a usable count. Studies that report counting error against a manual ground truth, and that compare instance-level against semantic-level modelling on the same images, are scarce.

Third, and most consequential for the credibility of any result in this area, evaluation protocols are frequently not matched to the structure of the data. Small microscopy datasets are hierarchical – many images per specimen, many augmented copies per image – and a partition that ignores this structure allows the same specimen, or the same source image, to appear on both sides of the split. Reported scores are then optimistic by an unknown margin.

This thesis addresses these gaps by treating bead quantification as an instance segmentation problem, by evaluating it with the counting and sizing errors that determine laboratory usefulness alongside conventional segmentation metrics, and by adopting a specimen-wise protocol whose effect relative to the naive alternative is measured rather than assumed.

2.6 Summary

Bead defects are the standard diagnostic of a misconfigured electrospinning process, and they are currently quantified by hand because existing automated tools target the fibre network instead. Classical thresholding fails on these micrographs because beads and fibres are not separable by intensity. Encoder–decoder networks segment the bead class well but merge touching instances, which corrupts precisely the quantity of interest. Region-based instance segmentation separates instances by construction and is the approach adopted here. The following chapter sets out the dataset, the calibration and the evaluation protocol in detail.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter defines the experimental design. It begins with the dataset, because its size and structure constrain everything that follows: 11 annotated micrographs drawn from four specimens is a small and strongly hierarchical sample, and the evaluation protocol has to be built around that fact rather than around convention. The chapter then describes the physical calibration that lets pixel measurements be reported in micrometres, the three methods under comparison, and the metrics.

3.2 Research Design

The study is experimental and comparative. Three segmentation methods – a classical thresholding pipeline, a semantic encoder–decoder network and a region-based instance segmentation network – are applied to the same micrographs under an identical partitioning scheme, and are assessed both by conventional segmentation metrics and by the counting and sizing errors that determine whether the output is usable in the laboratory.

The comparison is deliberately structured so that each method isolates one factor. The classical pipeline establishes what is achievable without training data and represents current practice. The semantic network establishes what learning buys while still producing only a foreground mask. The instance network adds explicit instance separation. The difference between the second and third therefore measures the value of instance-level modelling specifically, rather than the value of deep learning in general.

3.3 Dataset

3.3.1 Acquisition and Structure

The dataset consists of SEM micrographs of electrospun nanofibre mats, acquired on a Zeiss instrument at an accelerating voltage of 15.00 kV and working distances between 8.5 and 11.0 mm. Every micrograph is 1024×768 pixels, 8-bit greyscale, stored as JPEG.

The 11 micrographs are not independent samples. They derive from four specimens, denoted Z2, Z4, Z5 and Z6, each imaged at three magnifications: $500\times$, $1000\times$ and $3000\times$. The numeric suffix in a filename encodes the magnification rather than a repeat acquisition, so that Z2-1, Z2-2 and Z2-3 are three views of one specimen. One combination is absent – specimen Z4 has no $500\times$ micrograph – and specimen Z6 uses the identifier Z6-4 for its $3000\times$ view. Table 3.1 summarises the resulting layout.

Table 3.1. The 11 annotated micrographs. Specimen and magnification are recovered from the filename and confirmed against the instrument banner burned into each image. All micrographs are 1024×768 px at 15.00 kV.

Image	Specimen	Mag.	WD (mm)	Beads	Median d (px)
Z2-1	Z2	500×	10.0	71	14.2
Z2-2	Z2	1000×	10.0	51	28.8
Z2-3	Z2	3000×	10.0	10	76.3
Z4-2	Z4	1000×	9.0	59	24.4
Z4-3	Z4	3000×	9.0	4	68.0
Z5-1	Z5	500×	8.5	102	12.9
Z5-2	Z5	1000×	8.5	57	22.1
Z5-3	Z5	3000×	8.5	14	53.9
Z6-1	Z6	500×	11.0	152	15.7
Z6-2	Z6	1000×	11.0	62	28.7
Z6-4	Z6	3000×	11.0	24	75.4
Total	4 specimens	3 mags		**606**	

Figure 3.1 shows one specimen at all three magnifications with its annotations overlaid. The change in apparent object size is the dominant visual difference between the three panels, and it is the reason a scale-robust model is required.

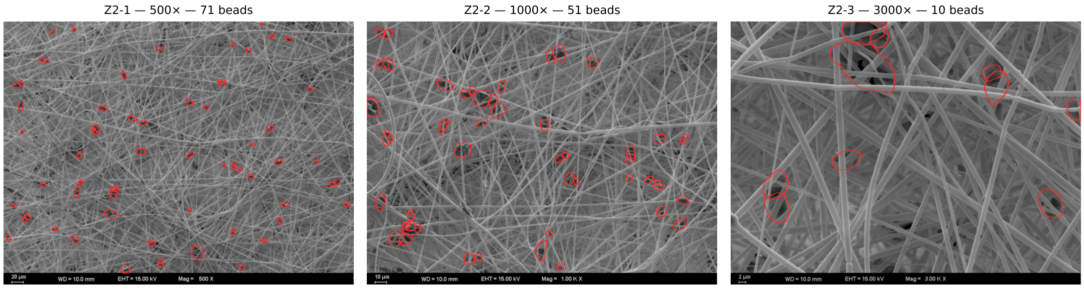

Figure 3.1. Specimen Z2 imaged at 500×, 1000× and 3000×, with the manual bead annotations overlaid in red. The same physical beads appear at roughly 14, 29 and 76 pixels across respectively, and the number of beads within the field falls from 71 to 10 as the field of view narrows. A model trained on these data must handle a sixfold range in object scale.

3.3.2 Annotation

Annotations were produced in LabelMe (version 5.4.1) and are supplied as one JSON file per micrograph. Every annotated object is a closed polygon carrying the single class label `bead`; polygons contain between 12 and 33 vertices, with a median of 20. Across the 11 micrographs there are 606 annotated beads. No annotation intersects the instrument banner region described in Section 3.3.4.

The dataset is therefore single-class: the task is to separate beads from everything else, and no distinction is drawn between bead subtypes or between beads and locally thickened fibre segments.

3.3.3 Physical Calibration

Each micrograph carries a burned-in scale bar. Measuring the bar in pixels and dividing by its stated physical length yields the pixel size, and the values recovered this way are mutually consistent: the field width is 284.4 μm at 1000× and scales inversely with magnification, as it must. The resulting calibration is given in Table 3.2.

The calibration can be validated independently of the scale bars. If it is correct, the physical bead size distribution measured at one magnification should agree with that measured at another, since all three views image the same specimens. Converting every annotated polygon to an equivalent circular diameter and grouping by magnification gives median diameters of 7.92, 7.34 and 6.75 μm at 500×, 1000× and 3000× respectively – agreement to within 15%, which satisfies the criterion set in Section 1.4 and confirms both the calibration and the internal consistency of the annotation.

Table 3.2. Pixel size recovered from the burned-in scale bars. The field width is inversely proportional to magnification, confirming that the three values form a consistent calibration rather than three independent measurements.

Magnification	Field width (μm)	Pixel size (nm/px)	Beads
500×	568.9	555.6	325
1000×	284.4	277.8	229
3000×	94.8	92.6	52

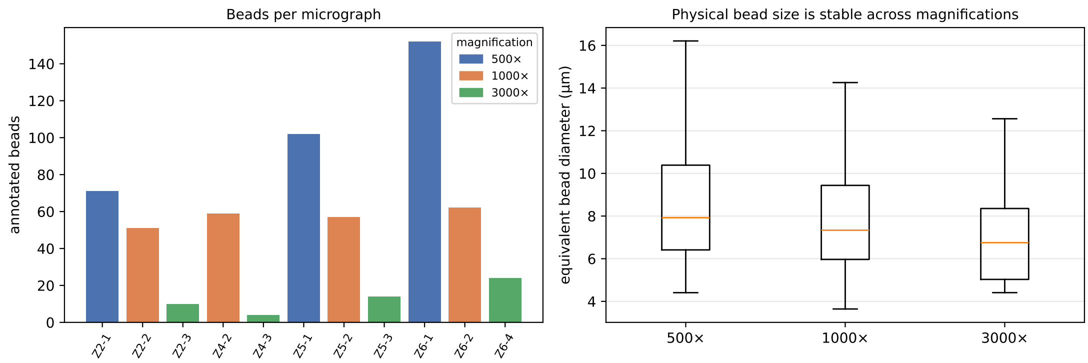

Figure 3.2. Left: annotated beads per micrograph, coloured by magnification. The count falls steeply with magnification because the field of view narrows, so the 3000× micrographs contribute only 52 of the 606 annotated beads. Right: distribution of equivalent bead diameter in micrometres, grouped by magnification. The three distributions overlap, which is the expected result if the scale calibration is correct, since all three magnifications image the same four specimens.

The residual downward trend with magnification – 7.92 to 6.75 μm – is consistent with a sampling effect rather than a calibration error. At 3000$\times$ the field is only 94.8 μm wide, so a large bead is more likely to be clipped by the image border and excluded, and only 52 beads are observed in total. This is discussed again when the results are interpreted.

3.3.4 The Instrument Banner

Every micrograph carries a burned-in information banner occupying the bottom 32 pixels, containing the scale bar, working distance, accelerating voltage and magnification. The banner is not part of the specimen. It is high-contrast, it is identical in structure across images, and it correlates with magnification, so a network trained on uncropped images could in principle read the magnification off the banner instead of learning scale-invariant features.

The banner is therefore cropped before any processing, leaving an effective image size of 1024×736. No annotation is lost by doing so, since no annotated polygon extends into the banner region. The calibration values in Table 3.2 are extracted from the banner once, before cropping, and carried forward as metadata.

3.3.5 The Supplied Augmentations

The dataset as received contains 143 image–annotation pairs rather than 11. Each of the 11 originals is accompanied by 12 pre-computed augmented copies: a horizontal flip, a vertical flip, eight translations of ± 150 pixels along each axis and in combination, and two horizontal stretches of 20% and 40%. The polygon coordinates were transformed along with the images, and inspection confirms that they remain correctly registered in every variant.

These copies are nevertheless excluded from the analysis in this thesis, for two reasons.

The first is statistical. Thirteen copies of one micrograph contain the information of one micrograph. Reporting 143 images, or 7 878 bead instances, would misstate the size of the sample by a factor of 13, and any metric computed over the augmented set would be dominated by repeated content.

The second is methodological, and more serious. Because the copies were generated before partitioning, a random split over the 143 files would place transformed versions of the same source micrograph in both the training and the test partition. The test score would then partly measure memorisation. Augmentation is still used in this thesis – it is essential at this sample size – but it is applied on the fly during training, after partitioning, so that no test image has a transformed relative in the training set. The augmentations actually used are described in Chapter 4.

Two artefacts in the supplied copies reinforce this decision. The flipped variants contain a mirrored instrument banner, in which the burned-in text is reversed, and the translated variants contain black padding along the edge vacated by the shift. Neither is a plausible micrograph, and both introduce structure that a network can exploit.

3.4 Evaluation Protocol

The four specimens, not the eleven micrographs, are the independent units of this dataset. Micrographs Z2-1, Z2-2 and Z2-3 are three views of the same physical material: they share its polymer, its spinning parameters and its particular population of beads. A partition that places Z2-1 in training and Z2-3 in testing is not measuring generalisation to a new specimen but generalisation to a new magnification of a specimen already seen.

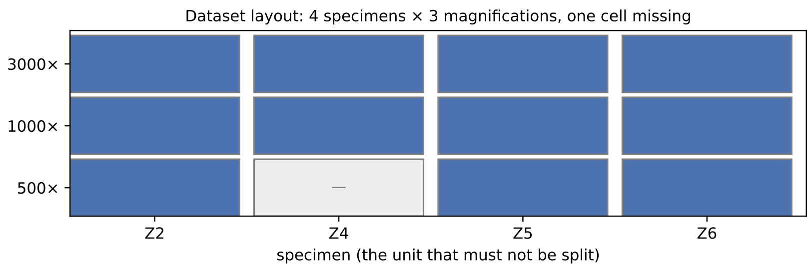

Figure 3.3. Layout of the dataset. Each column is a specimen and each row a magnification; the shaded cells are the 11 available micrographs. Partitioning must be performed along columns, because the specimen is the only independent unit. Specimen Z4 has no 500× micrograph.

The protocol adopted is therefore **leave-one-specimen-out cross-validation**. Each of the four folds trains on all micrographs of three specimens and tests on all micrographs of the fourth. Every micrograph is used for testing exactly once, and no specimen is ever split. With four specimens this gives four folds, and results are reported as the mean over folds together with the spread, which at this sample size is itself an important number to report.

To quantify what this protocol costs – and therefore what a naive protocol would have overstated – a second, deliberately flawed configuration is also run, in which the 11 micrographs are split at random without regard to specimen. The difference between the two is reported in Chapter 5. This comparison is one of the stated contributions of the thesis: it converts a methodological argument into a measured quantity.

3.5 Methods

3.5.1 Classical Baseline: Otsu Thresholding with Watershed

The classical pipeline represents current laboratory practice. The cropped micrograph is contrast-normalised, thresholded with Otsu's method [16], cleaned with morphological opening and closing to remove isolated fibre fragments, and the resulting binary mask is split into instances by a watershed transform [23] applied to its distance transform with local maxima as markers. Components smaller than a minimum area, expressed in physical units via the calibration of Section 3.3.3, are discarded.

This baseline requires no training data, so it is evaluated on all 11 micrographs directly. Its parameters are tuned once on the training folds and held fixed.

3.5.2 Semantic Baseline: U-Net

The semantic baseline is a U-Net [18] with a ResNet encoder [7] initialised from ImageNet weights [2]. It is trained to predict a binary bead-versus-background mask, and instances are recovered afterwards by connected-component labelling.

This baseline is included specifically to isolate the cost of merging. It is expected to segment bead pixels well while merging clusters of touching beads into single components; the gap between its pixel-level and its count-level performance is the quantity of interest.

3.5.3 Instance Segmentation: Mask R-CNN

The primary model is Mask R-CNN [6] with a ResNet-50 backbone [7] and a feature pyramid network [14], initialised from COCO-pretrained weights [15]. The region proposal network proposes candidate boxes, and for each surviving proposal the network predicts a class, a refined box and a binary mask within the region. Because each bead is handled in its own region, touching beads are separated by construction.

The feature pyramid is important for this dataset specifically: it assigns proposals to pyramid levels according to their size, which is the mechanism by which a single model can handle beads spanning 14 to 76 pixels across.

Anchor scales are set from the empirical bead size distribution rather than left at their COCO defaults, since bead defects are far smaller relative to the image than typical COCO objects – the median bead occupies 0.04% of the image area.

3.6 Evaluation Metrics

3.6.1 Segmentation and Detection Metrics

Detection and mask quality are reported as average precision at an intersection-over-union threshold of 0.5 (AP_{50}), at 0.75 (AP_{75}), and averaged over thresholds from 0.5 to 0.95 following the COCO protocol [15]. The aggregated Jaccard index [13] is reported additionally, because it penalises merges and splits explicitly rather than treating each object independently, and panoptic quality [11] is reported to separate recognition from delineation quality.

3.6.2 Laboratory Metrics

Segmentation metrics do not by themselves establish that a method is usable. The following are therefore reported as primary outcomes:

- **Counting error.** The signed and absolute relative error in the number of beads detected per micrograph, against the manual annotation.
- **Merge and split rates.** The proportion of ground-truth beads absorbed into a predicted instance containing another bead, and the proportion of single ground-truth beads divided across multiple predictions. These are the two failure modes

that corrupt a count, and they are reported separately because they have opposite effects on it.

- **Areal density.** Beads per square millimetre, computed from the detected count and the calibrated field area, which is the form in which the measurement is used for process comparison.

- **Size distribution agreement.** The predicted and manual distributions of equivalent bead diameter in micrometres, compared by their medians and interquartile ranges and by a two-sample Kolmogorov–Smirnov test.

3.7 Tools and Technologies

The implementation uses Python with PyTorch. Annotations are parsed from the LabelMe JSON format and converted to COCO-style instance masks. The classical baseline uses scikit-image and OpenCV. Figures and statistics are produced with Matplotlib and NumPy. Full version information is given in Chapter 4.

3.8 Summary

The dataset comprises 11 SEM micrographs of four electrospun specimens, imaged at three magnifications and annotated with 606 bead polygons. Each micrograph is calibrated against its burned-in scale bar, and the calibration is validated by the agreement of the physical bead size distribution across magnifications. Because the specimen is the only independent unit in the data, evaluation uses leave-one-specimen-out cross-validation, and the pre-computed augmented copies supplied with the dataset are excluded in favour of augmentation applied after partitioning. Three methods – classical thresholding, semantic U-Net and Mask R-CNN – are compared under this protocol, assessed both by segmentation metrics and by the counting and sizing errors that determine laboratory usefulness. The next chapter describes how this design is realised in software.

CHAPTER 4

IMPLEMENTATION

4.1 Introduction

This chapter describes how the design of Chapter 3 is realised in software: how the LabelMe annotations are converted into a trainable format, how the instrument banner and the physical calibration are handled, how augmentation is applied after partitioning, and how the three methods are configured and trained.

4.2 System Architecture

The pipeline is organised as five stages, each independently runnable so that the expensive stage – training – does not have to be repeated when only reporting changes.

1. **Ingest.** Read the LabelMe JSON files, recover specimen and magnification for each micrograph, and extract the pixel calibration from the instrument banner.

2. **Prepare.** Crop the banner, rasterise the polygons into instance masks, and emit a COCO-format annotation file.

3. **Partition.** Assign micrographs to the four leave-one-specimen-out folds.

4. **Train and predict.** Fit each method on each fold's training partition and predict on its held-out specimen.

5. **Evaluate and report.** Compute segmentation metrics, counting and merge/split statistics, and convert detections into physical units.

4.3 Data Preparation

4.3.1 Annotation Parsing and Rasterisation

Each LabelMe file stores a list of shapes, each with a class label, a shape type and a list of vertices. All 606 shapes in this dataset are polygons carrying the label `bead`. Polygons are rasterised into per-instance binary masks at full resolution. Object area is computed analytically from the polygon vertices using the shoelace formula rather than by counting mask pixels, which avoids a rasterisation bias that would be significant for the smallest beads: at 500$\times$ the median bead is only about 14 pixels across, so a one-pixel boundary error changes its area by roughly 15%.

Equivalent circular diameter is derived from the polygon area A as $d = 2\sqrt{A/\pi}$, and converted to micrometres using the magnification-specific pixel size from Table 3.2.

4.3.2 Banner Handling and Calibration

The instrument banner occupies the bottom 32 rows of every micrograph. Calibration is performed before cropping: the scale bar is located as the longest run of saturated pixels in the lower-left region of the banner, and its pixel length divided by the physical length printed beside it gives the pixel size. The recovered values are cross-checked by verifying that field width scales inversely with magnification, as reported in Section 3.3.3.

The banner rows are then removed, giving a 1024×736 working image. Polygon coordinates require no adjustment because the crop is taken from the bottom edge and no annotation extends into the banner region.

4.3.3 Fold Assignment

Fold membership is derived from the specimen prefix of the filename, so that all micrographs of one specimen move together. The four folds are Z2, Z4, Z5 and Z6. The assignment is written to disk as an explicit manifest rather than recomputed at training time, so that every method is guaranteed to see identical partitions and the comparison between them is exact.

The deliberately flawed control configuration described in Section 3.4 uses the same machinery with the specimen constraint disabled, splitting the 11 micrographs at random.

4.4 Augmentation

Augmentation is applied on the fly during training, after partitioning, for the reasons set out in Section 3.3.5. The pre-computed copies shipped with the dataset are not used.

The transformations are chosen to be label-preserving for this modality. Micrographs of a non-woven mat have no canonical orientation, so horizontal and vertical flips and rotations in multiples of 90° are physically meaningful. Random scaling is included deliberately and with a wider range than would be usual, because scale robustness across the 500× to 3000× span is the central difficulty of the dataset. Brightness and contrast jitter model variation in detector gain between sessions. Elastic and shear deformations are excluded: they would distort bead shape, and bead shape is a measured output rather than an incidental property.

Random cropping to fixed-size tiles is used both as augmentation and as a practical measure, since a 1024×736 micrograph containing up to 152 small beads is awkward to fit in memory at the batch sizes needed for stable training.

4.5 Model Configuration

4.5.1 Mask R-CNN

The primary model uses a ResNet-50 backbone with a feature pyramid network, initialised from COCO-pretrained weights. Two departures from the defaults are required by the data.

Anchor sizes are set from the empirical bead size distribution rather than left at their COCO values. The median bead occupies 0.04% of the image area, far smaller than a typical COCO object, and default anchors would leave the smallest beads without a well-matched proposal.

The maximum number of detections per image is raised well above the COCO default of 100, because a single 500× micrograph contains up to 152 beads and the default would silently truncate the count – an error that would corrupt precisely the quantity being measured.

4.5.2 U-Net Baseline

The semantic baseline uses a ResNet encoder initialised from ImageNet weights and a symmetric decoder with skip connections, trained with a combined cross-entropy and Dice loss to counter the strong foreground/background imbalance, since bead pixels are a small fraction of every micrograph. Instances are recovered from the predicted mask by connected-component labelling, with no boundary-aware post-processing, so that the merge behaviour this baseline is included to demonstrate remains visible rather than masked.

4.5.3 Classical Baseline

The classical pipeline is implemented with scikit-image: CLAHE contrast normalisation, Otsu thresholding, morphological opening and closing, distance transform, local-maximum markers and watershed. Its three free parameters – the morphological kernel size, the minimum accepted object area in μm^2 and the marker suppression distance – are tuned by grid search on each fold's training partition and then held fixed for that fold's test specimen, so that it is subject to the same protocol as the learned methods.

4.6 Training Setup

Table 4.1 lists the training configuration. Each of the four folds is trained independently from the same initialisation, and no fold's held-out specimen influences any choice made during its training.

4.7 Development Environment

Development is in Python 3.11 with PyTorch. Descriptive statistics and figures use NumPy, Matplotlib and Pillow; the classical baseline uses scikit-image and

Table 4.1. Planned training configuration, to be confirmed against the runs actually performed.

Setting	Mask R-CNN	U-Net
Backbone	ResNet-50 + FPN	ResNet-34 encoder
Initialisation	COCO	ImageNet
Optimiser	SGD, momentum 0.9	Adam [10]
Learning rate	*TBD*	*TBD*
Batch size	*TBD*	*TBD*
Epochs	*TBD*	*TBD*
Loss	RPN + box + mask	Cross-entropy + Dice

OpenCV.

4.8 Challenges and Solutions

Three problems were encountered during data preparation. Each is recorded here because each changed a design decision.

The augmented copies could not be used as supplied. The dataset arrives as 143 image–annotation pairs, which initially appears to be a usable sample size. The filenames reveal that these are 11 originals with 12 transformed copies each. Verifying the copies by overlaying their polygons on their images confirmed that the transformations are correctly applied, so the copies are not defective – but they are not independent either, and allowing them to cross a partition boundary would have produced optimistic results. The resolution was to discard them and augment after partitioning.

The banner is augmented along with the image. Inspecting the flipped copies showed the instrument banner mirrored, with the burned-in text reversed, and the translated copies contain black padding where the image was shifted away from the edge. This confirmed that the banner had not been accounted for during augmentation, and reinforced the decision to crop it before any processing.

Object size varies sixfold across the dataset. The median bead is 14 pixels across at 500× and 76 at 3000×. This drove the choice of a feature pyramid backbone, the resetting of anchor scales, and the wide random-scaling range in the augmentation policy.

4.9 Code Organization

- `code/dataset_stats.py` – annotation parsing, calibration, descriptive statistics and the dataset figures of Chapter 3.
- `code/prepare_data.py` – banner cropping, mask rasterisation and COCO conversion. *(to be written)*
- `code/folds.py` – specimen-wise fold manifest generation. *(to be written)*
- `code/train_maskrcnn.py`, `code/train_unet.py` and `code/classical.py` – the three methods. *(to be written)*
- `code/evaluate.py` – segmentation metrics, counting and merge/split statistics, and physical quantification. *(to be written)*

4.10 Summary

The pipeline separates ingest, preparation, partitioning, training and evaluation so that the partitioning guarantee of Chapter 3 is enforced in one place and inherited by every method. The two decisions that most affect the validity of the results – discarding the pre-computed augmentations and cropping the instrument banner – both follow from properties of the data discovered during preparation rather than from convention. The next chapter reports what the pipeline produces.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 Introduction

This chapter reports the experimental results. It first establishes the dataset characterisation and calibration, which are complete, and then reports the segmentation experiments under the leave-one-specimen-out protocol of Section 3.4.

5.2 Dataset Characterisation

The descriptive analysis of the dataset is complete and is reported in Chapter 3: 606 annotated beads across 11 micrographs of four specimens, at three magnifications, with the pixel calibration of Table 3.2.

The first objective of Section 1.4 is met. The median equivalent bead diameter is 7.92 μm at 500$\times$, 7.34 μm at 1000$\times$ and 6.75 μm at 3000$\times$, an agreement of within 15% against a stated criterion of 20%. Since the three magnifications image the same four specimens, this agreement is evidence both that the scale calibration is correct and that the annotation is internally consistent across magnifications – an annotator who applied a different size threshold at different magnifications would have broken it.

The residual downward trend with magnification is consistent with border clipping rather than calibration error. At 3000$\times$ the field is 94.8 μm wide, so a bead of 8 μm occupies a much larger fraction of it and is correspondingly more likely to intersect the image boundary and be excluded from annotation. Only 52 of the 606 beads come from 3000$\times$ micrographs, so this group is also the noisiest.

5.3 Experimental Setup

5.3.1 Training Configuration

5.3.2 Fold Composition

Table 5.1. Leave-one-specimen-out fold composition. Bead counts are known from the annotation and are filled in; results columns are added once the runs complete.

Fold	Test specimen	Test micrographs	Test beads
1	Z2	3	132
2	Z4	2	63
3	Z5	3	173
4	Z6	3	238
Total		**11**	**606**

The folds are markedly unbalanced: specimen Z4 contributes only 63 beads across two micrographs, while Z6 contributes 238 across three. Per-fold results are therefore reported individually as well as averaged, since a mean over four folds of this size is not a stable summary on its own.

5.4 Segmentation Results

5.4.1 Mask R-CNN

Table 5.2. Mask R-CNN under leave-one-specimen-out cross-validation.

Test specimen	AP_{50}	AP_{75}	AP	AJI
Z2				
Z4				
Z5				
Z6				
Mean $\pm$ s.d.				

5.4.2 U-Net Semantic Baseline

5.4.3 Classical Baseline

5.4.4 Method Comparison

Table 5.3. The three methods under an identical protocol, averaged over the four leave-one-specimen-out folds.

Method	AP_{50}	AJI	Counting error (%)	Merge rate (%)
Otsu + watershed				
U-Net + CC				
Mask R-CNN				

5.5 The Cost of a Naive Split

Table 5.4. The same model evaluated under a specimen-wise protocol and under a random split over the 11 micrographs. The difference is the amount by which a naive protocol would have overstated performance on this dataset.

Protocol	AP_{50}	AJI	Counting error (%)
Random split over images			
Leave-one-specimen-out			
Difference			

5.6 Behaviour Across Magnification

5.7 Physical Quantification

5.7.1 Bead Count and Areal Density

5.7.2 Size Distribution Agreement

5.8 Discussion

5.8.1 Key Findings

5.8.2 Comparison with the Literature

5.8.3 Limitations

The limitations below follow from the dataset and are known already; they should be stated regardless of how the experiments turn out.

Four specimens. The strongest claim this dataset can support is generalisation across four specimens of what appears to be one material system. It cannot establish generalisation to a different polymer, a different solvent or a different instrument.

Fifty-two beads at 3000×. The highest magnification is represented by only 52 annotated objects, so per-magnification results for 3000× rest on a small sample and should be read as indicative.

One annotator. All 606 annotations come from a single source, so annotator bias cannot be distinguished from ground truth, and no inter-annotator agreement can be reported.

Single class. The annotation does not distinguish beads from locally thickened fibre segments, so the boundary between the two is defined implicitly by the annotator's judgement rather than by a stated criterion.

5.8.4 Reliability of the Reported Numbers

5.9 Summary

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Conclusion

This thesis addressed the automated quantification of bead defects in SEM micrographs of electrospun nanofibre mats, a measurement currently performed by hand and therefore slow and difficult to reproduce.

The dataset work is complete and is reported in Chapter 3. Eleven micrographs of four specimens, imaged at three magnifications and carrying 606 manually delineated beads, were characterised and placed on a physical scale by recovering the pixel size from each micrograph's burned-in scale bar. The calibration was validated against the data themselves: the median equivalent bead diameter agrees to within 15% across the three magnifications, which is the expected result only if both the calibration and the annotation are consistent.

The analysis also established two properties of the dataset that determine how it must be used. The 143 image–annotation pairs supplied are 11 originals and 132 transformed copies, so the effective sample size is 11 and the copies cannot be allowed to cross a partition boundary. And the eleven micrographs derive from only four specimens, so the specimen rather than the image is the unit of partitioning. Both findings are reflected in the leave-one-specimen-out protocol adopted throughout.

6.2 Research Contributions

1. A characterised and physically calibrated dataset of 606 bead defects across 11 SEM micrographs of four electrospun specimens, with per-magnification pixel sizes recovered from the instrument's own scale bars and validated for internal consistency.

2. An evaluation protocol matched to the hierarchical structure of small microscopy datasets, in which the specimen is the unit of partitioning, together with a measurement of how much a naive split inflates apparent performance on this data.

3. A controlled comparison of instance segmentation, semantic segmentation and classical thresholding on identical micrographs under an identical protocol.

4. An open implementation covering annotation conversion, scale calibration, training, evaluation, and the reporting of bead count, areal density and size distribution in physical units.

6.3 Limitations

The limitations are set out in Section 5.9.3 and are summarised here: four specimens of a single material system; only 52 annotated beads at the highest magnification; a single annotator, so annotator bias cannot be separated from ground truth; and a single class, so the distinction between a bead and a locally thickened fibre segment rests on the annotator's implicit judgement rather than a stated criterion.

6.4 Future Work

6.4.1 Short-term

Extend the specimen count. The single most valuable addition to this work is more specimens rather than more images of the same ones. Four specimens support four cross-validation folds; ten would support a materially stronger claim, and the annotation cost per specimen is already known from this work.

Measure inter-annotator agreement. Having a second annotator re-delineate a subset would convert the largest stated limitation into a measured quantity, and would establish the ceiling against which any model should be judged. A model that matches inter-annotator agreement has reached the useful limit of the task.

Use the unannotated micrographs. If further micrographs exist without annotations, they can be used for self-training or consistency-based semi-supervised learning, both of which are well suited to a regime with abundant images and scarce labels.

Shape-prior architectures. Bead defects are approximately ellipsoidal and convex, which is exactly the assumption behind StarDist [20] and the flow-based representation of Cellpose [22]. Both are plausible alternatives to Mask R-CNN on this data and would be a natural extension of the comparison.

6.4.2 Long-term

Joint fibre and bead analysis. Bead quantification is one half of the morphological characterisation of a nanofibre mat; fibre diameter distribution is the other, and tools such as DiameterJ [8] address it separately. A single model reporting both from one micrograph would cover the characterisation a laboratory actually needs.

From measurement to process feedback. If bead statistics could be linked to the spinning parameters that produced them, the pipeline would move from characterisation towards prediction – estimating, from a micrograph, which parameter is out of range. This requires micrographs labelled with their process conditions, which the present dataset does not carry.

Annotation assistance from foundation models. The Segment Anything Model [12] cannot recognise beads on its own, but prompted with a detector's boxes it could refine mask boundaries, or accelerate the annotation of new specimens by turning a click into a polygon. Given that annotation effort is the binding constraint on this problem, this is the intervention with the highest leverage.

6.5 Final Remarks

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APPENDIX A

SOURCE CODE

This appendix lists the core implementation. The complete repository, including the training and evaluation scripts, is available at `[repository URL]`.

A.1 Dataset Characterisation and Calibration

The following module parses the LabelMe annotations, recovers the physical pixel size from each micrograph's burned-in scale bar, and produces the descriptive statistics and figures reported in Chapter 3. Object area is computed analytically from the polygon vertices rather than from a rasterised mask, for the reason given in Section 4.3.1.

```
# The 11 original micrographs, keyed by specimen and magnification. The
# suffix encodes magnification, not a repeat: -1 is 500x, -2 is 1kx,
# -3/-4 is 3kx.
IMAGES = {
    "Z2-1": ("Z2", 500), "Z2-2": ("Z2", 1000), "Z2-3": ("Z2", 3000),
    "Z4-2": ("Z4", 1000), "Z4-3": ("Z4", 3000),
    "Z5-1": ("Z5", 500), "Z5-2": ("Z5", 1000), "Z5-3": ("Z5", 3000),
    "Z6-1": ("Z6", 500), "Z6-2": ("Z6", 1000), "Z6-4": ("Z6", 3000),
}

# Calibrated from the burned-in scale bars: the field width is 284.4 um
# at 1000x and scales inversely with magnification, over 1024 px.
```

```python
NM_PER_PX = {500: 555.6, 1000: 277.8, 3000: 92.6}

BANNER_H = 32  # burned-in Zeiss info bar at the bottom of every micrograph


def polygon_area(points):
    """Shoelace area of a closed polygon, in px^2."""
    n = len(points)
    return abs(
        sum(
            points[i][0] * points[(i + 1) % n][1]
            - points[(i + 1) % n][0] * points[i][1]
            for i in range(n)
        )
    ) / 2.0


def collect():
    """Per-image bead areas, equivalent diameters and physical sizes."""
    out = {}
    for name, (specimen, mag) in IMAGES.items():
        ann = json.loads((DATA / "Labels" / f"{name}.json").read_text())
        areas_px = [polygon_area(s["points"]) for s in ann["shapes"]]
        diam_px = [2 * math.sqrt(a / math.pi) for a in areas_px]
        scale = NM_PER_PX[mag] / 1000.0  # um per px
        out[name] = {
            "specimen": specimen,
            "mag": mag,
            "n": len(areas_px),
            "diam_px": np.array(diam_px),
            "diam_um": np.array(diam_px) * scale,
```

```
            "area_um2": np.array(areas_px) * scale**2,
        }
    return out
```

Listing A.1. Polygon area and physical calibration (`code/dataset_stats.py`)

A.2 Verification of the Supplied Augmentations

The check reported in Section 3.3.5, confirming that the polygons of the pre-computed augmented copies remain correctly registered to their transformed images. The copies are correct but are not independent samples, and are therefore excluded from the analysis.

```
for name in ["Z2-1", "Z2-1_shift-150-0", "Z2-1_stretch-h-0.2
   ", "Z2-1_flip-h"]:
    im = Image.open(f"Images/{name}.jpg").convert("RGB")
    ann = json.loads(open(f"Labels/{name}.json").read())
    draw = ImageDraw.Draw(im)
    for s in ann["shapes"]:
        draw.line(
            [tuple(p) for p in s["points"]] + [tuple(s["
               points"][0])],
            fill=(255, 0, 0), width=3,
        )
    im.save(f"check_{name}.png")
```

Listing A.2. Overlaying annotations on augmented copies